Σ^0 Production in the NuMI Beam at MicroBooNE: Analysis Note

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Abstract

Working note for the MicroBooNE search for Σ^0 production in $\bar{\nu}_\mu$ charged-current interactions in the NuMI beam, $\Sigma^0 \to \Lambda\gamma$, $\Lambda \to p\pi^-$, with the full reverse-horn-current (RHC) exposure of 1.15×10^{21} POT. Two selections are maintained. The main one runs on standard PeLEE ntuples, which carry data, beam-off (EXT) and dirt samples, and ends in a cross-validated gradient BDT: $S = 2.12$ NuWro signal events on $B = 342$ (0.62% purity). The second follows the MicroBooNE Λ CCQE analysis on HyperonProduction ntuples, fitting the Λ decay vertex and testing the connectedness of the raw wire activity: $S = 0.90$ GENIE signal events on $B = 4.4$ (17% purity), with no data or dirt in that format yet. An expected-sensitivity calculation puts both under one signal model. Neither can observe Σ^0 at the predicted rate, so the result will be an upper limit. The best median expected 90% CL limit is about 2.7× the GENIE rate and 17× the NuWro rate; the two generators differ by a factor of six, because NuWro includes hyperon final-state interactions. With its BDT cut set for the background systematic, the PeLEE chain gives the better limit under both generators at every systematic up to 30%. The background systematic of the PeLEE chain is therefore the next piece of work. Revised 2026-10-19: the PeLEE background samples are complete (RHC dirt, the Run 3 overlay, EXT), the BDT is retrained with EXT, the Λ-style chain and its EXT estimate are documented, and the expected sensitivity is new (Secs. 2–8). Revised 2026-10-30: a dedicated GENIE Σ^0 QE sample replaces the GENIE hyperon sample, which turned out to hold only 754 distinct interactions (Sec. 2.4). Later on 2026-10-30: GENIE's $\bar{\nu}_\mu$ hyperon cross section is wrong in its hyperon-mass terms, Σ^0 1.85× too high and Λ 0.80× too low, while NuWro agrees with an independent calculation (Sec. 2.6). The GENIE yields and limits quoted in the tables predate the correcting weights, and the GENIE limits are correspondingly optimistic; the FHC GENIE sample will be weighted the same way (Sec. 12.1).

Contents

1 Status and summary **2**

2 Samples and normalisation **4**
2.1 The run3b NuWro signal production . . . 4
2.2 The NuMI flux weight . . . 5
2.3 Changes to the PeLEE background samples (2026-10-17) . . . 5
2.4 The GENIE Σ^0 QE sample (2026-10-30) . . . 5
2.5 Why the samples disagree on the absolute Σ^0 rate . . . 6
2.6 GENIE's hyperon cross section (2026-10-30) . . . 7

3 The PeLEE selection **9**
3.1 Signal definition and fiducial volume . . . 9
3.2 Candidates and cut chain . . . 10
3.3 BDT . . . 10
3.4 Results . . . 11
3.5 GENIE signal through the same selection . . . 12

3.6 Detector variations . . . 13

4 What limits the PeLEE purity **13**
4.1 The photon candidate . . . 14
4.2 Methods taken from the single-photon analyses . . . 14

5 Reconstruction study on HyperonProduction ntuples **14**
5.1 Why the PeLEE ntuples cannot answer these questions . . . 14
5.2 Signal definition used in this section . . . 15
5.3 Per-particle reconstruction . . . 15
5.4 Track/shower misclassification . . . 15
5.5 The pion candidate . . . 15
5.6 The Λ flight distance from the primary vertex is not measurable . . . 15

6 The Λ-style selection **16**
6.1 Motivation . . . 16
6.2 What it does differently . . . 16
6.3 The island test . . . 16
6.4 Operating point . . . 17

7 Beam-off background in the Λ-style chain **19**

8 Expected sensitivity **19**
8.1 Method . . . 19
8.2 Results . . . 20

9 A global hypothesis fit **21**
9.1 Motivation . . . 21
9.2 Measured ingredients . . . 21

10 Fake data by Poisson throw **22**

11 Framework port **22**

12 Exposure: combined FHC+RHC later **22**
12.1 The FHC GENIE signal sample: weights, not a fixed GENIE . . . 23

13 Work plan **23**

1 Status and summary

The analysis sets a limit on Σ^0 production in $\bar{\nu}_\mu$ charged-current interactions. The signature is

$$\bar{\nu}_\mu + \mathrm{Ar} \to \mu^+ + \Sigma^0 + X, \qquad \Sigma^0 \to \Lambda\gamma, \qquad \Lambda \to p\,\pi^-, \tag{1}$$

so the reconstructed final state is a muon, a proton, a pion and a photon of about 75 MeV. All yields in this note are scaled to the full RHC data exposure, 1.1526×10^{21} POT.

Two selections exist, on two ntuple formats, and they are fully independent (Table 1). The PeLEE chain (`Sigma/pelee/SigmaSelection.C`, Sec. 3) is the analysis: it is the only one that can see data, EXT and dirt. The Λ-style chain (`SigmaLambdaSel.C`, Sec. 6) needs the raw wire activity, which only the HyperonProduction ("hypana") format stores, and that format exists for MC and EXT but not for data or dirt.

Table 1: The two selections at their operating points. GENIE yields are from the GENIE Σ^0 QE sample with the PPFX CV weight, for both chains (Sec. 2.4). Efficiencies are quoted against each chain's own truth definition and are not comparable across columns; the signal yields under one model are. μ_{UL} is the median expected 90% CL limit on the rate in units of the model (Sec. 8).

	PeLEE BDT	Λ-style
ntuple format	PeLEE	HyperonProduction
data, EXT, dirt	all three	EXT only (factorised)
final discriminant	BDTG > 0.779	island test on ≥ 2 planes
S, NuWro (with hyperon FSI)	2.12	0.142
S, GENIE (no hyperon FSI)	13.5	0.84
B	342	4.40
purity, NuWro / GENIE	0.62% / 3.8%	3.1% / 16%
μ_{UL} at $f = 15\%$, NuWro / GENIE	43 / 6.8 (re-optimised 28 / 4.5)	38 / 6.3

What changed on 2026-10-30.

- **GENIE Σ^0 QE sample** (Sec. 2.4): 250 555 events, 1.23×10^{24} POT, in both formats. Its rate agrees with the plain overlay's own Σ^0 to 2%. It replaces the GENIE hyperon sample, which holds only 754 distinct Σ^0 interactions, each overlaid about 20 times, at 1.32× the overlay's rate. The Λ-style GENIE yield falls from 1.33 to 0.90, and the GENIE signal of the PeLEE chain now rests on 15 304 raw events instead of 187.
- **PeLEE BDT retrained** with clamped inputs and the GENIE signal added to the training (Sec. 3.3): 12% less background at 40–50% signal efficiency; the operating point moves to $0.759 \to 0.779$, S $2.11 \to 2.12$, B $382 \to 342$, and the expected limits improve by 5–10%.
- **The chain comparison changes under GENIE.** With the PeLEE cut set for the systematic, the PeLEE chain is ahead of the Λ-style chain under both generators up to $f = 30\%$; before, GENIE put the Λ-style chain ahead above $f \approx 20\%$.
- **Fake data** in the PeLEE chain is now the GENIE sample, a second generator rather than the NuWro signal itself.

What changed on 2026-10-19 (from the 2026-10-15 version).

- **PeLEE background samples completed** (Sec. 2.3). The dirt sample was FHC and is replaced by RHC dirt; the Run 3 overlay, 55% of the overlay statistics, was missing; the run3b beam-off (EXT) sample is added; and GENIE signal events inside the inclusive overlay are no longer counted as background.
- **BDT retrained** with EXT in the background tree and on 95% of each cross-validation fold rather than the 50% TMVA uses by default (Sec. 3.3). The figure of merit rises by 9.7% and the background at the operating point halves.
- **Detector variations** measured on the PeLEE chain: $\lesssim 7\%$ on the background, unresolved on the signal (Sec. 3.6).
- **The Λ-style chain** is documented here for the first time (Sec. 6). Its background now rests on 7.0×10^{21} POT of hypana overlay, and the photon conversion gap is part of its operating point.
- **EXT for the Λ-style chain**: a hypana EXT production and a factorised estimate, 0.62 ± 0.27 events (Sec. 7).

- **Expected sensitivity** for both chains under one signal model (Sec. 8). This reverses the earlier conclusion, drawn from purity and from $S/\sqrt{S+B}$, that the Λ-style chain is the stronger one.

What this analysis produces. Not a differential cross section and not an observation. The best median discovery significance at the predicted rate is 0.6σ (GENIE) and 0.1σ (NuWro), so the deliverable is a cross-section *upper limit*. The PeLEE ntuples carry the multisim weights the background covariance needs (`weightsGenie`, `weightsPPFX`, `weightsReint`), so this no longer depends on the framework port (Sec. 11).

2 Samples and normalisation

Table 2: Samples, scaled to 1.1526×10^{21} POT. The NuWro and dirt samples are on the old NuMI flux and carry a per-event flux weight (mean 0.677 for NuWro).

format	sample	content	scale
PeLEE	Σ^0 signal, other hyperons	NuWro run3b overlay, 249 832 events, 1.3394×10^{24} POT	8.61×10^{-4}
	GENIE Σ^0, fake data	GENIE Σ^0 QE overlay, 250 555 events, 1.2300×10^{24} POT	9.37×10^{-4}
	ν background	RHC overlay Runs 1, 2, 3, 4a, 4b, 4c, 1.5463×10^{22} POT	0.0745
	dirt	RHC Runs 1, 2b, 3b, 4a, 4b, 1.7409×10^{21} POT	0.662
	beam-off (EXT)	run3b, 3.265×10^7 gates	0.750
hypana	Σ^0 signal (GENIE)	`sig_genie_qe.list`, the QE sample above, 2506 files	9.37×10^{-4}
	Σ^0 signal (NuWro)	the run3b production above	8.61×10^{-4}
	ν background	`bkg_overlay.list`, 14 164 files, 7.0488×10^{21} POT	0.1635
	beam-off (EXT)	16 570 events from 3 349 run3b files, 162 912 gates	150.25
	dirt, data	none	

2.1 The run3b NuWro signal production

The Run 1-EXT sample `prod_hyperon_sigmazero_reco1_prodnuwro_mcc9` was retired on 2026-10-15. Its replacement is produced locally (`~/MicroBooNE/production/nuwro_prod/`): 2500 jobs of 100 NuWro Σ^0 events overlaid on swizzled NuMI **run3b ext_unbiased** data, through the full gen→reco2 chain, then both hypana and PeLEE ntuples.

	retired (Run 1 EXT)	run3b
files / events	2981 / 64716	2500 / 249832
generated POT	3.4141×10^{23}	1.3394×10^{24}
true Σ^0 CC $\bar{\nu}_\mu$	17714 (27.4%)	58545 (23.5%)
raw signal at preselection	3226	10931

The hypana `ana/MetaTree` sum and the merged PeLEE `SubRun` sum agree on the POT to four digits, and `check_ext_unique.py` confirms 250,000 distinct EXT events with none used twice. The cosmics come from run3b, a period the background overlay covers. Selection efficiency was unchanged by the swap (50.1% → 50.8% at preselection).

2.2 The NuMI flux weight

NuWro's flux files were converted from the *old* NuMI gsimple flux (g4numi v6, bugged beamline geometry), while the overlay and the data are on the current `g4_10_4` flux. Every NuWro event therefore carries the official MicroBooNE "New/Old" ratio weight, binned in $(E_\nu, \theta_{\text{beam}})$. The mean weight is 0.6774, and 0.6770 for the Σ^0 CC subset, flat to 0.1% even though those events sit 0.13 GeV higher in energy. The only cost is a wider spread (rms 0.297 against 0.226): 49.9k effective events of 58.5k. `SigmaSelection.C` computes the weight from `nu_e` and `true_nu_p*`; the hypana macros read a per-event table. The RHC dirt turned out to be on the old flux too (Sec. 2.3) and takes the same map.

2.3 Changes to the PeLEE background samples (2026-10-17)

- **The dirt was FHC.** The file used as "Run 1 dirt" (4.21×10^{20} POT) has the true-E_ν spectra of the FHC dirt samples ($\chi^2/\text{ndf} \approx 1$) and not of the RHC ones (χ^2/ndf 90–660): $\langle E \rangle$ for ν_μ is 1.60 GeV against 2.15 for RHC, and for $\bar{\nu}_\mu$ 2.28 against 1.72. It is replaced by the RHC dirt of Runs 1, 2b, 3b, 4a and 4b (1.7409×10^{21} POT).

- **The RHC dirt is on the old NuMI flux.** Its `ppfx_cv` profile in true E_ν matches the old-flux overlay in every bin ($\bar{\nu}_\mu$ 0.886–0.910 from 0 to 5 GeV) and not the current-flux overlay (0.904–1.012), so it takes the NuWro flux map.

- **The Run 3 overlay was missing**: five files, 5.5185×10^{21} POT, +55% background statistics.

- **EXT added.** Run3b beam-off, 3.265×10^7 gates, scaled by 2.4477×10^7 beam-on triggers (the trigger table's 23 534 937 per 1.10823×10^{21} POT, taken at 1.1526×10^{21}): 0.7497. Data files have *empty* `backtracked_*` vectors whenever there are PFPs, and the guard against ragged vectors had been dropping every neutrino-slice EXT event; it no longer requires them for data. The 1 655 repeated (run, subrun, event) in the file are dropped.

- **Overlay signal removed from the background.** The inclusive overlay contains 1 038 raw GENIE Σ^0 signal events in the true fiducial volume, and they had been counted in B while the NuWro sample models the signal. They are now kept out of B and counted separately as the GENIE signal prediction (Sec. 3.5).

2.4 The GENIE Σ^0 QE sample (2026-10-30)

The GENIE twin of the NuWro run3b production: GENIE `LambdaCCQE` (Pais $\Delta S = 1$ quasi-elastic), tune G18_10a_02_11a, the official dk2nu RHC flux on the current geometry (no flux weight), a primary Σ^0 selected at generation, and the same overlay chain on run3b EXT with no EXT event shared with the NuWro sample. 2506 jobs of 100 events, 1.23005×10^{24} POT (4.9×10^{18} per Σ^0); every job finished. It carries the truth muon momentum, so the 0.15 GeV/c muon threshold of the signal definition applies to it.

sample (hypana truth)	in-FV Σ^0	non-QE	χ^2/ndf vs QE: Q^2, p_μ, p_Σ
GENIE Σ^0 QE sample	$\mathbf{81.85 \pm 0.28}$	0	–
plain RHC overlay (510 raw)	83.39 ± 3.69	0.8%	19/14, 23/21, 39/19
previous GENIE hyperon sample	110.13 ± 1.46	3.0%	864/27, 1229/37, 724/31
NuWro (FSI, flux-weighted)	12.82 ± 0.09	–	3250/34, 628/39, 4223/37

The QE sample agrees with the overlay's own Σ^0 in rate (0.98 ± 0.04) and in Q^2, p_μ and Λ flight; p_Σ differs by 2.5σ. The overlay's non-QE Σ^0 come with a kaon, which the signal definition vetoes, so the QE sample is the complete GENIE prediction for this signal.

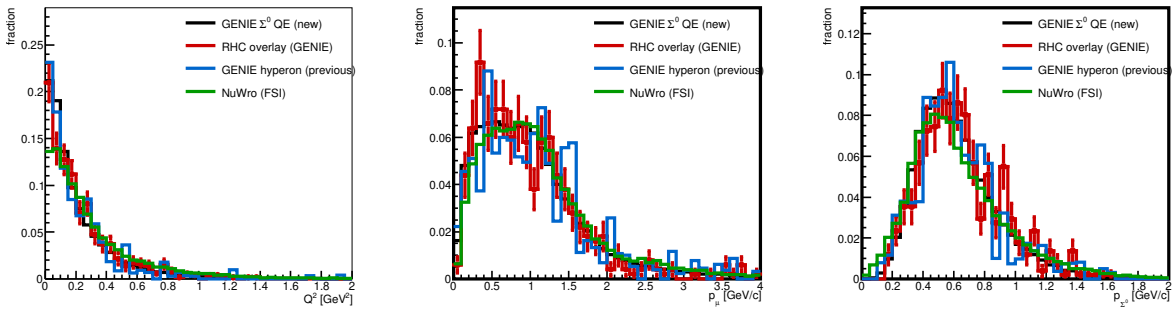


Figure 1: Truth Q^2, muon momentum and Σ^0 momentum of in-FV $\bar{\nu}_\mu$ CC Σ^0, area-normalised: the GENIE Σ^0 QE sample, the plain RHC overlay, the previous GENIE hyperon sample (754 distinct interactions) and NuWro.

PPFX CV weight. The PeLEE overlay carries `ppfx_cv`; a sample made from generator files has no flux-parent record. `PpfxMap.C` measures the overlay's $\langle$`ppfx_cv`$\rangle$ against true E_ν for $\bar{\nu}_\mu$ CC in the FV (0.90 at 0.6 GeV rising to 1.00 at 3 GeV). On the overlay's own Σ^0 the map predicts 0.934 against 0.938 measured, and `SigmaSelection.C` applies it to the QE sample. The PeLEE chain then agrees with the overlay's signal to 2% at every cut (Sec. 3.5). The Λ-style GENIE yield is multiplied by 0.934 as well; its background, like all hypana MC, carries no PPFX weight.

The previous GENIE hyperon sample. `sig_genie.list` holds 15 341 $\bar{\nu}_\mu$ Σ^0 events, but only **754 distinct interactions**, each overlaid about 20 times on different cosmics (identical lepton momentum, hyperon momentum and Q^2; `DupGen.C`). Its spectra are spiky, its statistics overstate the information about 20×, every parity split on it (hypana BDT folds, island tuning cross-validation) had copies of the same interactions on both sides, and its rate is 1.32× the overlay's with a harder muon spectrum ($\langle p_\mu \rangle$ 1.08 against 0.99 GeV/c). Everything that used it as the GENIE signal has been redone with the QE sample.

2.5 Why the samples disagree on the absolute Σ^0 rate

On one truth definition and one fiducial volume, Σ^0 CC $\bar{\nu}_\mu$ events per 1.1526×10^{21} POT with the true vertex in the FV:

sample	in-FV Σ^0	vs the plain overlay
plain RHC overlay (unfiltered GENIE)	83.3	1.00
GENIE hyperon sample `sig_genie`	110.1	1.32
GENIE Σ^0 QE sample (2026-10-30)	81.9	0.98
NuWro run3b, POT scale only	18.9	0.23
NuWro run3b, with the flux weight	12.8	**0.15**

The fiducial fractions are 37–38% in all three, so this is not a generation-volume artefact.

It is hyperon FSI. The NuWro configuration (`ultrapure-sigma0-numi-rhc.txt`) sets `dyn_hyp_cc=1` with every other channel off and, within it, `hyp_lambda=0`, `hyp_sigma_zero=1`, `hyp_sigma_minus=0`: NuWro generated the Σ^0 channel *and nothing else*, so its quoted POT per event is the exposure per Σ^0 *produced*. Everything else in the sample is final-state interaction: of the $\bar{\nu}_\mu$ CC events only 25.3% still carry a Σ^0, 27.9% carry a Λ ($\Sigma^0 N \to \Lambda N$ in the cascade), 10% a $\Sigma^\pm$, and 37.3% no strange particle at all. The MicroBooNE Λ paper (arXiv:2212.07888) states the other half outright: "the NuWro predictions include the effect of final state interactions, while

GENIE does not take them into account". For GENIE, produced = surviving, so the comparison that means anything is Σ^0 *produced*:

	in-FV Σ^0 produced per full-RHC POT
GENIE overlay (no hyperon FSI)	83.3
NuWro, POT scale only	79.1
NuWro, with the flux weight	53.6

The 6.5× gap decomposes exactly as 4.19× (FSI) ×1.48× (flux weight) ×1.05× (production rate), product 6.51× against the measured 6.51×. The sign flip between channels is the same effect: in Fig. 2 of the Λ paper NuWro sits *above* GENIE ($\simeq 2.1$ against $\simeq 1.5 \times 10^{-40}$ cm^2/Ar) because FSI *adds* Λ, and far below for Σ^0 because it *removes* Σ^0.

A control run verifies the normalisation. The same flux, target and configuration with `dyn_hyp_cc=0 dyn_qel_cc=1` – ordinary CC quasi-elastic instead of the hyperon channel – gives 7.539×10^{15} POT per event over NuWro's box. The overlay's own CC QEL rate (20,277 in-FV events in 7.1479×10^{20} POT) converts to 8.835×10^{15} on the same footing, using the 37.0% box-mass fiducial fraction and the 0.677 flux factor. NuWro's generic event rate is therefore within 17% of GENIE's, the ordinary CCQE model difference, and its flux × target normalisation is sound. Referenced to each generator's own CCQE rate, NuWro's Σ^0 *production* is 0.81× GENIE's.

What this means for the limit. A Σ^0 yield taken from the NuWro sample is a yield *with* hyperon FSI; one taken from the GENIE samples is a yield *without* it. The difference is roughly a factor of four and it is a model choice, not a bug. The measured Λ cross section, $2.0^{+2.2}_{-1.7} \times 10^{-40}$ cm^2/Ar, brackets both generators and cannot distinguish them. Every result in this note is therefore quoted under both models. The hyperon sample's 1.32× above belonged to that sample: the QE sample agrees with the overlay (Sec. 2.4). The 1.61× that the Λ cross-check of Sec. 6 finds against the published GENIE prediction was measured on the hyperon sample and has to be redone. Both chains now take GENIE from the QE sample.

Superseded in part (2026-10-30). The "production" comparison above compares NuWro with GENIE as if GENIE's Σ^0 cross section were right. It is not: GENIE's Σ^0 is 1.85× too high, and once that is corrected the two generators agree on Σ^0 production to 12% (Sec. 2.6). The FSI factor stands.

2.6 GENIE's hyperon cross section (2026-10-30)

Generators with the hyperon cascade off. NuWro was rerun with no cascade and no jailing (`kaskada_on=0`, `hyp_Eb=0`), one hyperon channel at a time, on the configuration of the signal production and with the flux weight; GENIE is taken from the generator files of the QE sample (Σ^0) and of the same production with Λ kept (`genie_to_hepmc.C`, `keep_pdg`). In-FV $\bar{\nu}_\mu$ CC events with a primary hyperon per 1.1526×10^{21} POT (`GenRate.C`):

	NuWro	GENIE	NuWro/GENIE
Λ	223.7 ± 2.9	209.5 ± 1.4	1.07
Σ^0	47.9 ± 0.6	81.7 ± 0.6	0.59
Σ^-	115 ± 2	≈ 197	0.58
Σ^0/Λ	0.214	0.390	

NuWro's $\Sigma^-/\Sigma^0 = 2.40$ is the isospin factor 2 times argon's N/Z (2.44), and GENIE's is too. Against E_ν the NuWro/GENIE ratio rises the same way for Λ and Σ^0 (Fig. 2, centre); only the normalisation differs, by 0.55 at every energy. So the flux weight is not the cause, and the difference is in the Σ couplings or the formula, not in the flux or the nucleus.

Same couplings. Both generators use the Cabibbo SU(3) form factors with the same coefficients ($f_1^\Lambda = -\sqrt{3/2}\,F_1^p$, $g_1^\Lambda = -(1+2x)g_A/\sqrt{6}$, $f_1^{\Sigma^0} = -(F_1^p + 2F_1^n)/\sqrt{2}$, $g_1^{\Sigma^0} = (1-2x)g_A/\sqrt{2}$, and $\Sigma^- \sqrt{2}\times\ \Sigma^0$), $x = F/(F+D) = 0.365$ and $M_A = 0.99$–1.0 GeV (GENIE `LwlynSmithFF::Strange*`, NuWro `ff.cc`). NuWro contracts the full hadronic and leptonic tensors (`Singh_Model`); GENIE uses a closed form, `PaisQELLambdaPXSec`.

An independent calculation. `HypXsecWeights.py` evaluates $L_{\mu\nu}H^{\mu\nu}$ for $\bar\nu_\mu p \to \mu^+ Y$ from explicit Dirac traces with the same form factors. On ordinary CCQE it gives 1.06 (ν) and 0.41 ($\bar\nu_\mu$)$\times 10^{-38}$ cm^2 at 1 GeV, the Llewellyn Smith values. For hyperons, free proton, 10^{-40} cm^2:

	trace			GENIE spline		
E_ν	Λ	Σ^0	Σ^0/Λ	Λ	Σ^0	Σ^0/Λ
1 GeV	1.33	0.25	0.19	1.02	0.46	0.45
2 GeV	2.15	0.41	0.19	1.71	0.76	0.44

NuWro's 0.21 on argon agrees with the trace; GENIE's 0.45 is 2.4× higher.

Where GENIE goes wrong. A transcription of `PaisQELLambdaPXSec::XSec` (v3_00_06) reproduces GENIE's Λ spline (1.05 against 1.02 at 1 GeV; its Σ^0 is 15% under GENIE's, which has $G_E^n \neq 0$). With the final-state mass set to the nucleon's, it agrees with the trace to 0.5% and gives Llewellyn Smith CCQE. As the final-state mass is raised to the hyperon's, the two separate: the trace cross section falls, as phase space requires, while the Pais Σ^0 cross section *rises* (0.49 → 0.60 at 2 GeV, against 0.50 → 0.41). The error is therefore in the terms that depend on $M_Y - M_N$; relative to the trace, GENIE's Λ is ×0.80 and its Σ^0 and Σ^- ×1.5–1.85 (Fig. 2, left). Nothing in this analysis tests NuWro's formula beyond its agreement with the trace.

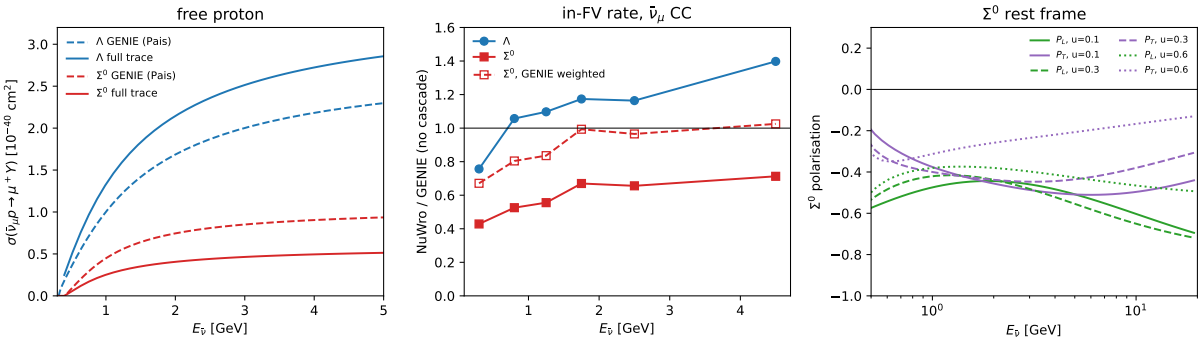


Figure 2: Left: $\bar\nu_\mu p \to \mu^+ Y$ on a free proton, GENIE's Pais splines (dashed) and the full trace with the same form factors (solid). Centre: NuWro/GENIE in-FV rate with the cascade off, flux-weighted, against true E_ν; open squares are GENIE Σ^0 with the cross-section weight. Right: Σ^0 polarisation in its rest frame against E_ν at three values of $u = (Q^2 - Q^2_{\min})/(Q^2_{\max} - Q^2_{\min})$.

Correcting the GENIE samples. A weight $w(E,u) = \left[d\sigma/dQ^2\right]_{\text{trace}} / \left[d\sigma/dQ^2\right]_{\text{Pais}}$ for a free nucleon, with E the $\bar\nu_\mu$ energy in the struck-nucleon frame and $u = (Q^2 - Q^2_{\min})/(Q^2_{\max} - Q^2_{\min})$ (`hyp_xsec_weights.root`, `HypXsecWeight.h`). For a two-body reaction on a given nucleon E and Q^2 fix the final state up to a flat azimuth, so the weight corrects the production kinematics as well as the rate, provided GENIE populates the phase space: its Pais expression turns negative and is clipped to zero only near $u \to 1$, which holds $\leq 0.3\%$ of the corrected cross section, and the weight is capped at 5 there. Σ^0 weights run from 0.63 to 1.4, Λ from 1.05 to 1.35 below the cap. On the generator files:

in-FV per data POT	GENIE	GENIE weighted	NuWro	NuWro/GENIE weighted
Λ	209.5	263.0	223.7	0.85
Σ^0	81.7	54.4	47.9	0.88
Σ^0/Λ	0.390	0.207	0.214	

The relative statistical error of the Σ^0 rate is 0.75% with the weight and 0.76% without, so it costs no statistics. Taking the struck nucleon at rest instead of from the event, as the PeLEE ntuples require, changes the rates by 1%. The common 0.85 is a nuclear (binding, hyperon potential, Pauli) and flux difference; the weighted shapes (Q^2, p_μ, p_Σ) move by at most 5% in their means. `SigmaSelection.C` applies the weight to every hyperon-QE event of every GENIE sample (overlay, dirt, detector variations, the QE sample), not to NuWro.

Σ^0 polarisation. The same traces with the spin projector $(1+\gamma_5\not{s})/2$ give the Σ^0 polarisation: P_L and P_T (in the production plane, towards the $\bar{\nu}_\mu$) are each about -0.4 to -0.5, and $P_N = 0$ without T-odd form factors (Fig. 2, right). The sign conventions are checked against explicit helicity spinors and on a point-like $V-A$ current, which gives $|\vec{P}| = 1$ at every angle. Neither generator chain polarises: GENIE and NuWro hand Geant4 an unpolarised Σ^0, which decays isotropically. The M1 decay passes the polarisation on as $\vec{P}_\Lambda = -(\vec{P}_\Sigma \cdot \hat{n})\hat{n}$, $\hat{n}$ the Λ direction in the Σ^0 frame, and $\Lambda \to N\pi$ distributes the nucleon as $1 + \alpha\vec{P}_\Lambda \cdot \hat{q}$ ($\alpha = 0.747$ for $p\pi^-$). `PolWeights.C` weights each event by $w = 1 - \alpha(\vec{P}_\Sigma \cdot \hat{n})(\hat{n} \cdot \hat{q})$, taking the production kinematics from the PeLEE ntuple and the Λ and nucleon directions from the hypana ntuple of the same production (start and end points). The rebuilt directions give 75.4 ± 10 MeV/c for the Λ in the Σ^0 frame (two-body value 74.5) and 108 ± 26 MeV/c for the nucleon in the Λ frame (100.6): the proton's start-to-end direction is approximate but usable. $\langle w \rangle = 0.997$ and its rms is 0.15 per event. Its effect on the selected yields and on the limits is being measured with the reweighted selection; Table 7 does not yet include either weight.

What the weights cannot correct. The hyperon's nuclear potential and the nucleon binding near threshold shift the hyperon kinetic energy rather than reweight it; GENIE has neither. Hyperon FSI changes the event topology: a weight can drop events with a survival probability but cannot produce the Λ from $\Sigma^0 N \to \Lambda N$. For NuWro the polarisation weight uses the post-cascade Σ^0 kinematics, which for the 24% that survive is the production kinematics only for those that did not scatter.

NuWro's cascade on Λ. For comparison with the Σ^0 numbers of Sec. 2.5: of the primary Λ, 64% leave the nucleus as Λ, 32% are jailed and 4% convert to Σ; the in-FV Λ rate falls to 0.72. Of the primary Σ^0 only 24% survive. The factor of about two in Σ^0 reduction quoted for NuWro in the second-class-current study is not reproduced by this build (`nuwro_old`, `hyp_Eb`=30 MeV); that is open.

3 The PeLEE selection

3.1 Signal definition and fiducial volume

`nu_pdg` $= -14$, `ccnc` $= 0$, a Σ^0 (PDG 3212) in the final-state list `mc_pdg`, a true muon momentum above 0.15 GeV/c, no other meson at the primary vertex, and the true vertex inside the fiducial volume. The PeLEE truth record stops at the primary vertex, so the decay is tagged by the Σ^0 itself, and no proton, pion or photon threshold can be imposed. The NuWro PeLEE file has no `mc_px`, so the muon threshold is *not applied* to the NuWro signal; it removes 4.2% of signal muons where it can be applied. The meson veto removes nothing from the NuWro sample.

Fiducial volume: $3.00 < x < 253.35$, $-112.53 < y < 114.47$, $3.1 < z < 1033.0$ cm, excluding $675.1 < z < 775.1$ cm. This is looser than the CC1μ1π box: the signal is rare enough that acceptance is worth more than vertex quality.

3.2 Candidates and cut chain

- **Muon**: longest track with `trk_score` > 0.8, start within 20 cm of the vertex, length > 10 cm, LLR PID > 0.7.
- **Proton and pion**: each non-muon track is scored for both roles with a quadratic logistic model and the best ordered pair is taken (`PidAssign.h`). Against HyperonProduction truth this raises the pion-candidate purity from 33.5% (ranking on a single discriminant) to 42.5%; the ceiling is 51.4%, the fraction of selected signal that contains a reconstructed true π^- at all. The candidates are formed when the tracks exist but are *not required*: demanding them removed more signal than background ($-31.6\%/-18.3\%$ and $-52.9\%/-43.5\%$) and halved the figure of merit. Events without a pair carry sentinels and the BDT input `HasPair`.
- **Photon**: the non-track PFPs closest to the vertex; the closest is the Σ^0 decay photon. Its energy, from the per-PFP `shr_energy_y_v`, averages 70 MeV in signal against the expected 74.5 MeV.

Nine cumulative stages: `AllEvents`, `FiducialV`, `CosmicVeto` (`topological_score` > 0.20), `NShowers` ($0 < n_{\rm shr} < 4$), `MuonCandidate`, `ProtonCandidate` and `PionCandidate` (pass-through, see above), `NTracksCut` ($n_{\rm trk} < 5$), `BDTG`. Derived observables: Λ and Σ^0 masses, Λ and photon vertex-pointing angles and impact parameters, Λ decay distance, a π^0 di-photon mass, and the second-shower block of Sec. 4.2.

3.3 BDT

Gradient BDT over 57 inputs, evaluated with two-fold cross-validation: folds are trained on even and odd event numbers and each is applied only to the events it did not train on. Every BDT number in this note is out of sample.

Retrained 2026-10-30 (current). Two changes. (1) Every input is mapped into its bulk range (`BDTClamp.h`, applied in `SigmaSelection.C` to the training trees and the reader alike); besides the cases below, the five pair distances carry a 10^5 "missing" value in 21–59% of events and `ShrSubClusters` an INT_MAX one in 29%, now capped just above their real maxima. (2) The GENIE Σ^0 QE signal (43 183 events) joins the NuWro signal in training, scaled to the same summed weight. Out of sample on NuWro:

training	AUC	B/B_0 at signal efficiency			peak $S/\sqrt{S+B}$	
		50%	40%	20%	value	cut
2026-10-18 (NuWro signal)	0.9000	0.0358	0.0215	0.0061	0.1077	0.759
clamped, NuWro signal	0.9057	0.0336	0.0203	0.0063	0.1102	
clamped + GENIE (adopted)	**0.9086**	**0.0317**	**0.0186**	**0.0055**	**0.1146**	**0.779**
clamped + GENIE, seed 1	0.9093	0.0312	0.0186	0.0055	0.1147	

On the GENIE signal the AUC rises from 0.9107 to 0.9204. More distinct signal helps even when the BDT is judged only on NuWro, as it did for the hypana BDT (Sec. 2.4).

Retrained 2026-10-18. Two changes, measured separately on the same events with EXT in B:

1. **EXT in the background tree.** The run3b beam-off events (8 977 raw, 25% of the training background weight) were not in the previous training.

2. **Train on 95% of each fold.** TMVA's default splits each fold into train and test halves; the test half feeds only TMVA's overtraining printout, so each fold trained on 25% of all events.

training	AUC	B/B_0 at signal efficiency			peak $S/\sqrt{S+B}$	
		50%	40%	20%	value	cut
2026-10-15 (no EXT, half fold)	0.8894	0.0412	0.0253	0.0067	0.0993	0.690
EXT, half fold (3 seeds)	0.8956–0.8966	0.0378–0.0385	0.0228–0.0236	0.0059–0.0062	0.1037–0.1049	
EXT, 95% of fold (adopted)	**0.9000**	**0.0358**	**0.0215**	**0.0061**	**0.1077**	**0.759**
EXT, 95% of fold (2 other seeds)	0.9002–0.9017	0.0352–0.0354	0.0209–0.0213	0.0059–0.0060	0.1084–0.1092	

EXT in the training gives +5% in the figure of merit, almost all of it EXT rejection: at 50% signal efficiency 0.9% of the EXT survives against 2.2%, with the overlay rate unchanged. Training on the full fold gives another +4%, and every full-fold seed beats every half-fold seed; reseeding moves the figure of merit by only ±0.5%. That the BDT still improves when it sees more signal is the signal-statistics limit again. The default seed is adopted, not the best one.

Input ranges (measured 2026-10-19, adopted 2026-10-30). Several inputs carry values far outside their physical range. `Shower1VtxDist` and `Shower1DeDxY` carry the −FLT_MAX "missing" sentinel (24% and 40% of background). The calorimetric energies of the pion and proton candidates have garbage up to 2×10^{33} (1.7% and 3.0% of background above 2 GeV). Five background events have +FLT_MAX in the Λ and Σ^0 masses. TMVA places its `nCuts` grid evenly between a node's minimum and maximum, so one such value leaves the variable unusable in that node; with bagging, almost every root node contains one of the five events and so cannot cut on the masses. Mapping each input into its bulk range, with distinct underflow and overflow values (`ClampBDTInputs.C`), and retraining gives AUC 0.9042 against 0.9006 (two seeds against three), −4% background at 50% signal efficiency, −2% at 40%, nothing at 20%, and +1.6% in the figure of merit. Every clamped seed beats every present seed, so the gain is real. That measurement missed the six sentinel variables named above, which the 2026-10-30 retraining also caps.

3.4 Results

Table 3: PeLEE chain at the operating point BDTG > 0.779. Efficiency is raw signal against true Σ^0 in the FV. B = other hyperons + overlay + dirt + EXT.

stage	raw signal	eff. [%]	S	B	purity [%]	$S/\sqrt{S+B}$
MuonCandidate	11107	51.61	6.469	28978	0.022	0.0380
NTracksCut	10931	50.79	6.370	26549	0.024	0.0391
BDTG	**3686**	**17.13**	**2.122**	**342.0**	**0.617**	**0.1144**

B at the BDTG point	BDT 2026-10-15, 0.616	2026-10-18, 0.759	current, 0.779	raw
other hyperons (NuWro)	1.74	1.06	1.08	1865
overlay	715.7	367.9	328.5	4642
dirt	13.8	2.26	1.93	6
EXT	109.5	11.25	10.50	14
total	840.7	382.4	342.0	
S	2.850	2.110	2.122	3686

The first column is the 2026-10-15 BDT with the completed samples, EXT included. At MuonCandidate EXT is 6 844 events against 20 348 from the overlay; after the BDT it is 3% of the background. Figure 3 shows the score.

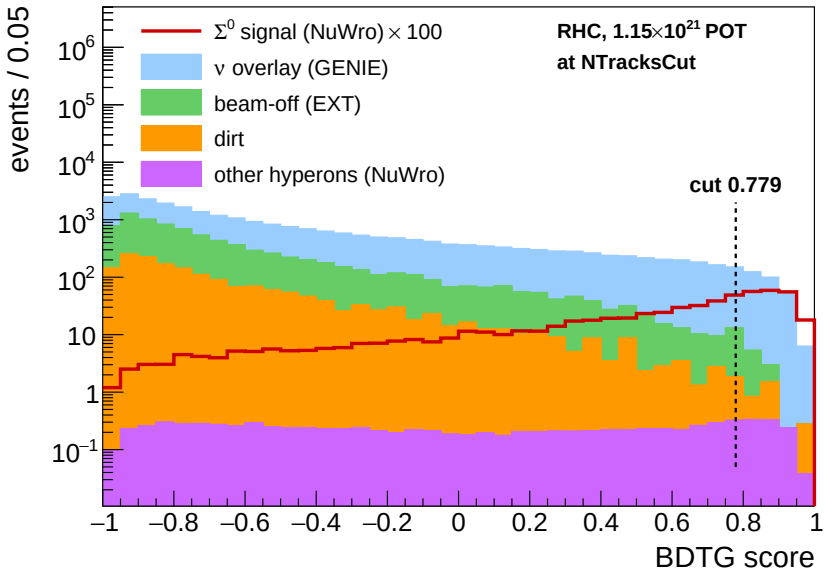


Figure 3: Cross-validated BDTG score at `NTracksCut`, background stacked by sample and the NuWro signal scaled by 100. The dashed line is the operating point.

The operating point should follow the systematic budget. 0.779 is the peak of $S/\sqrt{S+B}$. There a 2% background systematic is ±6.8 events against 2.1 signal, and the figure of merit cannot see it. Section 8 chooses the cut by the expected limit instead: 0.735 at $f = 0$, 0.885 at 10%, 0.915 at 15%.

3.5 GENIE signal through the same selection

The GENIE Σ^0 QE sample (Sec. 2.4) and the overlay's own GENIE Σ^0 events are both run through the full selection and BDT and counted apart from every sample, so the PeLEE chain has a GENIE signal yield under the same definition. Since 2026-10-30 the BDT trains on the QE sample's signal, with the same two-fold parity split, so its yield is still out of sample; the overlay's Σ^0 are never trained on, and the two agree to 2.4% after the BDT.

	QE sample			overlay Σ^0		
stage	raw	S_{GENIE}	eff. [%]	raw	S	GENIE/NuWro
AllEvents (true FV)	84089	73.61	100	1038	72.56	5.86
MuonCandidate	43760	38.39	52.2	553	38.67	5.93
NTracksCut	43183	37.88	51.5	544	38.08	5.95
BDTG	15304	13.47 ± 0.11	18.3	187	13.15 ± 1.02	6.34

The two GENIE yields agree to 2.4% or better at every cut. The efficiencies differ between generators: 18.3% for GENIE against 16.9% for NuWro (weighted). The limit on the number of signal events is therefore model-independent to 8%; what depends on the model is the rate it is compared with.

3.6 Detector variations

The Run 3 RHC detector-variation set re-simulates the same generated events under nine variations. `run_detvar.sh` runs the selection on each file and `DetVar.C` compares each variation with the CV on the events they share, with the CV weight on both sides. The samples are on the old flux, so only the ratios are used.

variation	S var/CV	B var/CV
LY attenuation	1.000 ± 0.000	1.005 ± 0.005
LY down	0.974 ± 0.026	0.991 ± 0.006
LY Rayleigh	0.974 ± 0.039	0.984 ± 0.009
recombination	1.125 ± 0.073	0.924 ± 0.030
SCE	0.904 ± 0.107	0.973 ± 0.035
WireMod θ_{XZ}	0.992 ± 0.078	0.970 ± 0.025
WireMod θ_{YZ}	0.902 ± 0.088	0.976 ± 0.027
WireMod X	0.956 ± 0.078	0.972 ± 0.027
WireMod YZ	0.947 ± 0.082	0.985 ± 0.026
quadrature (MC noise alone)	20.2% (21.4%)	9.6% (7.0%)

At the operating point 0.779. The signal is not resolved: its quadrature sum equals the MC noise. The background now is: recombination lowers it by $7.6 \pm 3.0\%$ (2.5σ), and the quadrature sum exceeds the noise by about 6.6% in quadrature. The CV keeps 51 raw signal and 1 174 raw background events, so the background detector systematic is about 7–10% ($\pm$24–33 events) and the signal one is not measured. At the previous point (0.759) the background sum was 5.2% on 6.8% noise.

4 What limits the PeLEE purity

The studies in this section were made on earlier versions of the chain (before the joint assignment, the 2026-10-18 retraining and the completed samples). The fractions carry over; the absolute yields do not.

Table 4: Surviving background by interaction mode, at a looser earlier operating point.

Mode	Scaled yield	% of background
Resonant (RES)	1381.8	52.3
Deep inelastic (DIS)	635.3	24.0
Quasi-elastic (QE)	363.8	13.8
Meson exchange (MEC)	177.1	6.7
Cosmic	74.6	2.8
Coherent (COH)	8.9	0.3
Total	2643.1	100

The surviving background is π^0 events. RES and DIS together are 76%. Both produce $\pi^0 \to \gamma\gamma$, which supplies a fake Σ^0 photon, plus charged hadrons that fake the proton and pion tracks. The background is not diffuse: it is one physics process.

The candidate assignment is the weak link. Truth-matched purities of the candidates in background events, before the joint assignment: muon 81%, proton 58%, pion 31% (another proton 28%), photon 29% (a proton 18%). The pion and photon candidates are wrong roughly

70% of the time, so both masses are built from misidentified objects in most background events. Section 5 measures the same on signal, on the hypana ntuples.

4.1 The photon candidate

Truth-matching the photon candidate in background (69 557 preselected events): photon 30.0%, unmatched 26.7%, charged pion 16.3%, proton 15.2%, muon 9.8%, electron 1.4%. Charged tracks reconstructed as showers are 41%.

The signal photon is soft, which inverts the strategy. The Σ^0–Λ mass splitting is only 77 MeV, so

$$E_\gamma = \frac{m_{\Sigma^0}^2 - m_\Lambda^2}{2m_{\Sigma^0}} = 74.5 \text{ MeV} \tag{2}$$

in the Σ^0 rest frame. The photon candidate averages 70 MeV in signal and 140 MeV in background, where π^0 photons dominate. Requiring shower-like quality would remove signal preferentially. Cuts in the soft direction help at preselection ($E_\gamma < 120$ MeV: 94.5% of signal, 25.4% of background removed, figure of merit $\times 1.09$) but not on top of the BDT ($\times 1.012$), which already uses the photon energy. An explicit π^0 mass veto rejects only 9% of background, because it needs *two* reconstructed showers. The shower Molière angle, the one unused shape variable, was added to the BDT and changed the figure of merit by $\times 1.013$.

4.2 Methods taken from the single-photon analyses

The gLEE single-photon search and the NCπ^0 technote isolate a single photon against π^0 events in which one decay photon was missed, which is our problem.

1. **Second-Shower Veto variables** – evidence of a photon that was not reconstructed: the unassociated-hit count and fraction in the slice (recomputed per event), `slclustfrac`, `secondshower_Y_nhit`, `secondshower_Y_dot` and the sub-cluster counts. *In the BDT*: they are among the 57 inputs.

2. **One BDT per background class** (π^0, non-π^0 neutrino, cosmic) instead of one BDT for everything. `TrainBDTSplit.C`, `apply_split.C` and `eval_split.C` implement it; it was run once on a 1 004-signal-event tree and has not been compared with the current single BDT. A split of the training by `HasPair` was measured and does not help.

3. **Conversion distance** as a first-class variable: realised in the Λ-style chain as the photon gap (Sec. 6).

4. **Constrain the π^0 rate from data.** The NCπ^0 technote finds GENIE over-predicts non-coherent NCπ^0 by 25%. Our background is the same physics, so its normalisation belongs in the covariance and, with data, in a sideband constraint (Sec. 13).

5 Reconstruction study on HyperonProduction ntuples

5.1 Why the PeLEE ntuples cannot answer these questions

Every question about *which particle a candidate actually is* is unanswerable on the PeLEE ntuples: `mc_pdg` lists the Σ^0 and the muon, but never the $\Sigma^0 \to \Lambda\gamma \to p\pi^-\gamma$ chain, and *zero of 17721 signal events* contain a backtracked π^- PFP. The HyperonProduction analyser writes the full decay chain together with per-PFP truth matching (`pfp_true_pdg`, `pfp_purity`, `pfp_completeness`). `HypInput.h` adapts the hypana branches to the PeLEE names the selection uses.

Three things to know about the format. It has no implicit neutrino-slice requirement: PeLEE events without a neutrino slice die at the fiducial cut, but hypana attaches a vertex to them

anyway, which inflated the background 2.4× until the explicit `flash_match_nu_slice_ID` ≥ 0 stage was added; the two chains then agree to 6–8% throughout. Its `trk_energy_proton` is the proton *momentum*, not the kinetic energy that the PeLEE branch of the same name holds; with the conversion the Λ mass width falls from 23.5 to 7.8 MeV (half inter-quartile range), and the PeLEE chain was never affected. And its `pfp_true_origin` is the analyser's own enum (2 is a Λ daughter), not `simb::Origin`. The measurements below were made on the retired NuWro sample; they are efficiencies and ratios, which the swap does not move.

5.2 Signal definition used in this section

Σ^0, $\bar{\nu}_\mu$ charged current, $\Lambda \to p\pi^-$ in truth, muon momentum above 0.15 GeV/c, and **both** the reconstructed and the true vertex inside the FV. Only $\sim 60\%$ of Σ^0 events have $\Lambda \to p\pi^-$ at all (PDG 63.9%). Requiring only the *reconstructed* vertex in the FV gives badly wrong answers: events whose reco vertex is in the FV but whose true vertex is not are mis-reconstructed, and they dominate. With reco FV only, the muon appears to be reconstructed 53.7% of the time and all four particles 10.2%; with both, 91.5% and 18.1%.

5.3 Per-particle reconstruction

particle	primary (gen 2)	secondary (gen ≥ 3)	not reconstructed	total
μ^+	91.5%	0.3%	8.2%	91.8%
p	58.1%	2.8%	39.1%	60.9%
π^-	45.7%	1.6%	52.8%	**47.3**%
γ	59.8%	2.6%	37.6%	62.4%

Table 5: Reconstruction of the four Σ^0 final-state particles (3704 signal events). The π^- is the weak link.

All four are found together in 18.1% of events, the hard ceiling for any selection that requires all four. Secondaries do not explain the loss: keeping generation-3 PFPs (new `pfp_generation` branch) recovers only 2.8% for the proton and 1.6% for the pion.

5.4 Track/shower misclassification

In events with at least four PFPs, with the selection's split (`pfp_trk_shr_score` > 0.5): the muon is a track 95.7% of the time, the proton 70.3%, the pion 54.2% (a shower 11.6%), and the photon is a *track* 15.0% of the time. All four are found in 41.2% of these events but correctly classified in only 26.7%. Photons passing as tracks compete for the pion slot, which is the origin of the 25% photon contamination among pion candidates.

5.5 The pion candidate

On signal reaching the pion stage, a true π^- PFP exists in only 56.6% of events, and 76.3% have exactly one track-like PFP left after the muon and proton are taken. The joint assignment of `PidAssign.h` (Sec. 3) is the improvement that was available; raising the pion track-score threshold lifts purity but lowers the figure of merit monotonically.

5.6 The Λ flight distance from the primary vertex is not measurable

Measured with truth-matched p and π^- tracks ($n = 266$): the true Λ flight has median 1.85 cm and mean 3.23 cm ($\beta\gamma \approx 0.4$, so $\beta\gamma c\tau \approx 3$ cm), the reconstructed primary-vertex resolution is 5.54 cm, and the reconstructed and true flight distances are uncorrelated ($r = -0.014$). Three

decay-vertex estimators give the same ~ 5.5 cm residual, all limited by the primary vertex; the tracks themselves converge well (DCA median 0.33 cm).

What this does not rule out. The measurement kills a cut on the distance between the Pandora vertex and the decay point. It does not kill the displaced topology: the Λ-style chain (Sec. 6) fits the decay vertex from the two tracks, requires the pair momentum to point back at the event vertex, and tests at wire level whether the pair is separated from the muon. That test passes 80% of signal with 2–3 cm flight and 98.6% of the signal it keeps has flown more than 2 cm.

6 The Λ-style selection

6.1 Motivation

The MicroBooNE Λ CCQE analysis (Thorpe & Nowak, Hyperon Technote v2) selects $\bar{\nu}_\mu$ CC Λ with 91% purity. At the same 6.7% efficiency the earlier hypana BDT chain for Σ^0 had 2.8%. Three causes were identified: our preselection rejected long-flight Λ (the pion score penalised tracks displaced from the primary vertex), there were no pair-level requirements, and the photon is an extra object. Reimplemented on our files (`LambdaSel.C`), their chain reproduces their efficiency to $0.79\times$ (our pair selector is provisional and their decay BDT is replaced by cuts), but the GENIE signal normalisation comes out $1.61\times$ theirs, which is unexplained and affects every Σ^0 yield. `SigmaLambdaSel.C` follows their chain.

6.2 What it does differently

- **Every ordered track pair is scored**, not each track separately. The pair selector's only topological input is the separation of the two track starts, which does not penalise a Λ that flew.
- **Pair precuts are a preference, not a veto**: as a veto they cost $\sim 20\%$ of signal at the same figure of merit.
- **The decay vertex is fitted** (a V fit, which recovers 3.78 cm for a true 4.00 cm flight), giving a decay length and the pointing angle α between the pair momentum and the vertex-to-decay direction.
- **The island test is a cut**, applied last.

6.3 The island test

A wire $\times$ time grid of the stored activity is built on each plane, clusters are seeded at the muon, proton and pion track starts, and the test requires the proton and pion clusters to merge while the muon stays separate (Fig. 4). The coordinate transforms were derived and validated for this (wire $= (Ay + Bz + C)/0.3$ reproduces 98.7% of the stored vertex wires within two wires; tick $= 18.1509\,x + 818.1$). The test reads raw wire windows, which exist only in the hypana format.

Connectivity is tuned. The original 3 wires $\times$ 40 ticks bridged the decay V to the muon whenever the Λ flew less than ~ 3 cm, because 40 ticks is 2.2 cm of drift. 2×20 lifts the signal pass rate:

true flight	1–2 cm	2–3 cm	3–5 cm	5–10 cm
3×40 (old)	19%	45%	70%	86%
2×20 (now)	43%	80%	85%	91%

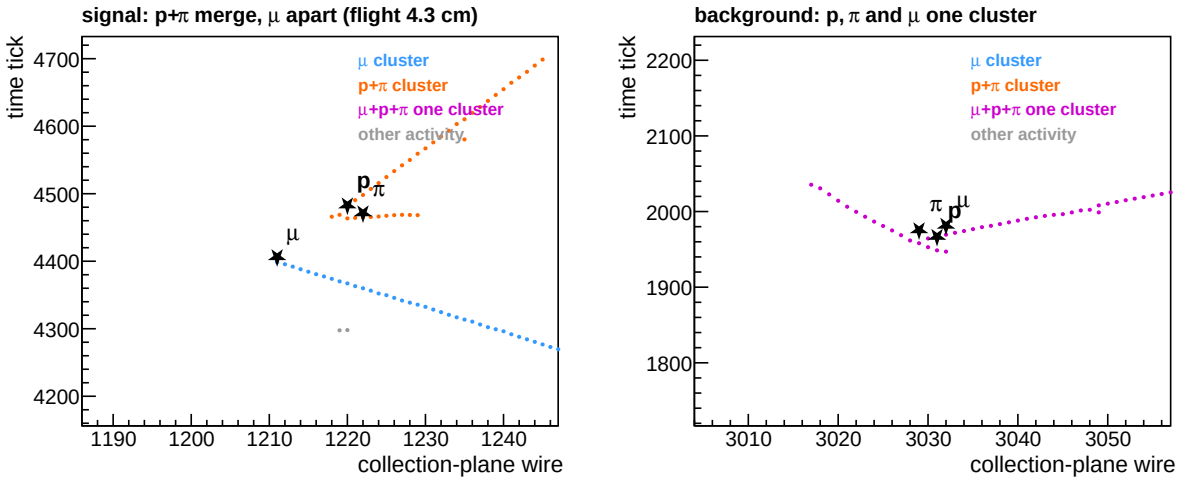


Figure 4: The island test on the collection plane. Left: a GENIE signal event with 4.3 cm true Λ flight; the proton and pion seeds share a cluster (orange) and the muon has its own (blue). Right: a background event at the same stage, whose proton, pion and muon seeds land in one cluster.

On one plane the figure of merit is flat in the connectivity; the gain is from requiring two planes (6.8× background rejection against 3.7×). The tuning used the Run 3 overlay list; re-scanned with the choice made on Run 3 and the score taken on the independent extension, the extension's figure of merit is a plateau (0.425–0.438) and 2×20 is 0.004 from its top. The two-plane pass rate is the same for both generators: 50.5% of NuWro, 50% of GENIE.

6.4 Operating point

Λ mass within ±20 MeV, decay length > 0.5 cm, $\cos\alpha > 0.95$, photon start > 2 cm from the vertex, island test passed on ≥ 2 planes. Background from the full hypana overlay; the GENIE signal column of Table 6 is the previous hyperon sample, on which the chain was developed:

Table 6: Λ-style cut flow. Errors are MC statistics.

stage	raw signal	raw background	S (GENIE)	B
AllEvents	15043	1840942	290.1	301031
SliceID	7801	776242	150.4	126931
MuonCandidate	3043	123000	58.7	20113
PairCandidate	921	51238	17.8	8378
Λ mass ±20 MeV	504	16391	9.72	2680
pointing ($\cos\alpha$, decay length)	144	706	2.78	115.4
photon gap > 2 cm	139	549	2.68	89.8
island, ≥ 2 planes	**69**	**23**	$\mathbf{1.33 \pm 0.16}$	$\mathbf{3.76 \pm 0.78}$

Absolute efficiency 0.46% (GENIE) and 0.42% (NuWro), purity 26% before EXT. 98.6% of the surviving signal has true Λ flight > 2 cm, against 43% in the BDT chain. With the NuWro signal the same point gives $S = 0.142 \pm 0.009$ (245 raw). NuWro's own non-signal events, Σ^0 that FSI turned into something else (mostly Λ), add 0.021 ± 0.003 background events (37 raw). With EXT (Sec. 7) $B = 4.40$.

With the GENIE Σ^0 QE sample the operating point gives $S = 0.896 \pm 0.029$ (956 raw; one plane: 1.343, 1433 raw), purity 17% with EXT, against 1.33 ± 0.16 (69 raw) from the hyperon sample. The normalisation accounts for a factor 0.74; the rest is a 10% lower efficiency (1.09%

of in-FV signal against 1.21%). The island parameters were tuned on the hyperon sample, whose few hundred interactions appeared about 20 times each, so part of that tuning may have fitted them; a retune on the QE sample is open.

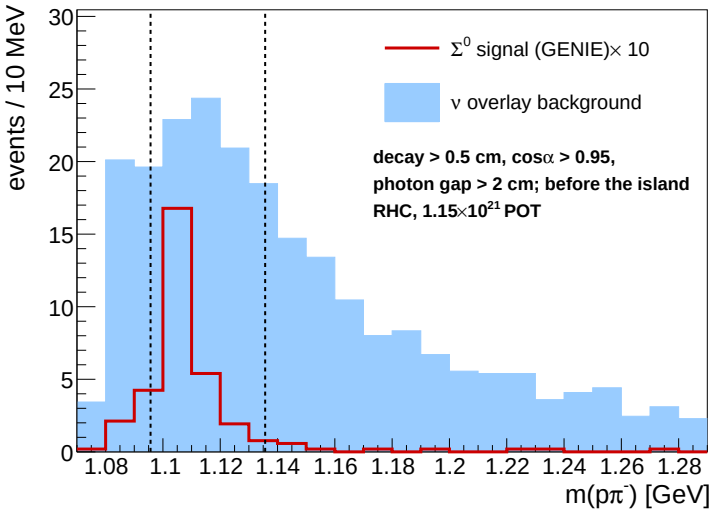


Figure 5: $p\pi^-$ mass after the pointing cut and the photon gap, before the mass window (dashed) and the island test. The signal peak sits slightly low because the pion momentum is taken from range under the muon hypothesis, which is 20% low for a pion but gives a narrower peak than any correction. The background is largely genuine $p\pi$ from Δ decay, whose mass sits just above threshold.

Out-of-sample checks. Without the gap, the original Run 3 list gives $B = 4.15 \pm 1.31$ (10 raw) and the independent extension 7.6 ± 1.4 (28 raw); every stage up to the pointing cut agrees per POT between the two to 1–4%, so the low Run 3 value is a fluctuation of the sample the island test was tuned on. The photon gap was fixed in September on the Run 3 list; on the extension it halves the background ($7.56 \to 4.05$) at 96% signal efficiency. Odd/even cross-validation chooses the same configuration in both folds.

The photon and Michel handles. 18.7% of background "photons" start within 2 cm of the vertex against 3.5% of signal; at two planes 11 of the 15 events the gap removes are hadrons reconstructed as a shower at the vertex. The threshold sits on a plateau from 1 to 8 cm. A Michel tag (a contained muon with another PFP within 5 cm of its end) exploits that the background muon is a wrong-sign μ^- 78% of the time and is captured 75% of the time in argon. It removes 87% true μ^-, but costs 22% of signal and lowers the figure of merit on top of the gap, so it is off.

Sidebands (one-plane point, for statistics):

region	S	B	raw B	S/B
signal region (blind)	1.89	14.9	91	12.7%
mass sideband, +40 to +200 MeV	0.02	10.5	64	0.2%
α sideband, $0 < \cos\alpha < 0.8$	0.50	58.2	356	0.9%
island-fail region	0.79	74.9	458	1.1%

All three are RES/DIS-dominated, like the signal region. The mass sideband exists only above the peak, because $m(p\pi)$ has a kinematic floor at 1.078 GeV.

7 Beam-off background in the Λ-style chain

The run3b `ext_unbiased` files were taken through reco2 and HyperonProduction locally: all 3 349 swizzled files, 16 570 NuMI-EXT-triggered events, every one of which reached the hypana ntuple. An unbiased event is one sampled gate, so the sample holds 162 912 gates and each event stands for 150.25 data events; it has 0.67% of the data's cosmic statistics.

That is too little to follow EXT to the operating point: none of the 76 MuonCandidate events reaches the pointing cut, and zero means < 346 events at 90% CL. `ExtFactor.C` therefore factorises: EXT at MuonCandidate from the hypana EXT, times the pass fraction to the operating point of the overlay's own cosmic-dominated candidates (more than half of the selected slice's hits unmatched to MC, i.e. overlaid cosmic data: 4 220 of the 123 000 cached events).

step (cumulative)	EXT raw	EXT P	MC cosmic raw	MC cosmic P	MC all-bkg P
MuonCandidate	76 ($11\,419 \pm 1\,310$)	–	4 220	–	–
pair	8	0.105 ± 0.035	571	0.135 ± 0.005	0.42
Λ mass ±20 MeV	3	0.38 ± 0.17	165	0.289 ± 0.019	0.32
pointing	0	0 of 3	7	0.042 ± 0.016	0.043
island + gap	0	–	0 of 7	–	0.0326

EXT at the operating point is 0.62 ± 0.27 events. The cosmic MC is used as far as the pointing cut, the last step with at least five cosmic events. For the island and gap steps the estimate takes the neutrino-background rate (3.3%); the cosmic MC alone gives 0 of 7 there, and without that assumption the bound is EXT < 6.2 at 90% CL. This is the open part of the estimate, and its direction is not known.

The real cosmic data test the factorisation wherever they still have events. 8 of 76 EXT candidates have a pair, which agrees with the cosmic MC (0.135, $P(\leq 8) = 0.29$) and excludes the neutrino-background rate (0.42 would predict 32, $P = 1.8 \times 10^{-9}$). Taking EXT's own counts instead of the cosmic MC fractions up to the pair step gives 0.48, and up to the mass window 0.62.

Hypana EXT at MuonCandidate is 1.7× the PeLEE beam-off at the PeLEE MuonCandidate, while the two chains' overlay agrees there (20 113 against 20 348). The excess is cosmic-specific and expected: hypana's `topological_score` is the highest NuScore of any slice, not the flash-matched slice's, which raises a cosmic-only event's score far more than a neutrino event's. The MC cosmics use the same definition, so the selection is consistent within hypana.

8 Expected sensitivity

8.1 Method

`Sensitivity.C` gives the expected 90% CL upper limit on μ, the Σ^0 rate in units of a generator's prediction ($\mu = 1$ is the model), for both chains under both generators. Each chain is a single-bin counting experiment, $n \sim \text{Pois}(\mu S e + b)$:

- b is lognormal with mean B and fractional width $\sqrt{r_{\text{MC}}^2 + f^2}$, where r_{MC} is the MC statistics of B (EXT and dirt included; a component with no raw events left is given one event of its own weight) and f is the background systematic, which is scanned;
- e is lognormal with mean 1 and fractional width $\sqrt{g^2 + r_S^2}$, where $g = 20\%$ is a placeholder for flux, efficiency and POT and r_S the signal MC statistics;
- the limit is Bayesian, with a flat prior in $\mu \geq 0$ and b, e marginalised. Hybrid CL_s is 4% tighter throughout, and both reproduce the PDG table exactly for a known background;

- the expected limit is the limit at the median n of Pois(B), with the band from the 16% and 84% quantiles;
- Z_A is the median discovery significance with an uncertain background (Cowan et al., arXiv:1007.1727).

$\mu_{\mathrm{UL}} = n_{\mathrm{UL}}(B, \sigma_b)/S$, so the generator only rescales the limit and the cut that minimises it does not depend on the signal normalisation.

8.2 Results

Table 7: Median expected 90% CL limit on μ, $g = 20\%$. "Re-optimised" scans the BDT cut from 0.50 to 0.95 for each f (cut in brackets); its GENIE value is the NuWro value divided by the GENIE/NuWro ratio measured at 0.779. BDT of 2026-10-30.

f	PeLEE, BDTG > 0.779		PeLEE, re-optimised		Λ-style	
	NuWro [16–84%]	GENIE	NuWro (cut)	GENIE	NuWro [16–84%]	GENIE
0%	17.2 [12.2, 24.9]	2.71	17.0 (0.735)	2.69	36.9 [25.6, 51.9]	6.22
5%	22.3	3.51	21.7 (0.855)	3.42	37.0	6.24
10%	32.4	5.11	24.9 (0.885)	3.92	37.2	6.28
15%	43.4 [37.0, 51.0]	6.84	28.3 (0.915)	4.47	37.6 [25.9, 52.7]	6.34
20%	54.1	8.52	31.5 (0.915)	4.97	38.1	6.42
30%	73.6 [66.2, 81.8]	11.59	36.3 (0.945)	5.72	39.3 [26.9, 54.8]	6.63

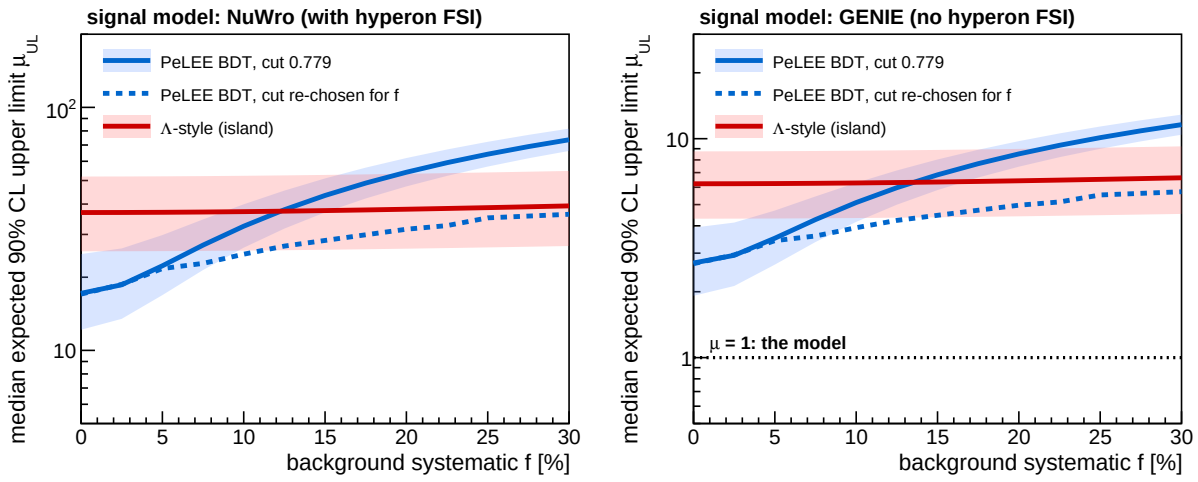


Figure 6: Median expected 90% CL limit on μ against the background systematic, under NuWro (left) and GENIE (right). Bands are the 16–84% range; the dotted line on the right is the GENIE prediction itself.

1. **The result will be a limit.** The best median significance at the model rate is 0.69σ (GENIE, PeLEE, $f = 0$) and 0.11σ under NuWro. The best expected limit, about 2.7× GENIE and 17× NuWro, is about 210 in-FV Σ^0 events in 1.15×10^{21} POT, against 74 predicted by GENIE and 13 by NuWro. In events the two models agree to 8% (216 and 199 at $f = 0$), because the efficiencies do (Sec. 3.5).
2. **The PeLEE chain gives the better limit under both generators up to $f = 30\%$.** With its cut set for the systematic it is 2.2× better than the Λ-style chain at $f = 0$, 1.5× at 10% and 1.3× at 15% (NuWro; GENIE 2.3×, 1.6×, 1.4×). At 30% the two are closest:

36.3 against 39.3 (NuWro), 5.72 against 6.63 (GENIE). Before 2026-10-30, GENIE put the Λ-style chain ahead above $f \approx 20\%$; that came from the hyperon sample's higher efficiency at the Λ-style point. The chains' GENIE/NuWro ratios are now 6.34 (PeLEE) and 5.89 (Λ-style). The Λ-style chain's purity is real, but its signal yield is 15× lower, and at $B \approx 4.4$ its limit is set by Poisson counting: removing all background uncertainty moves it by 3%. This reverses the conclusion of the earlier comparison by $S/\sqrt{S + B + (fB)^2}$, which set the Λ-style chain against the older hypana BDT, used the GENIE hyperon-sample normalisation, and kept each cut where it had been chosen at $f = 0$.

3. **The background systematic drives the PeLEE result.** From $f = 0$ to 15% its limit grows 1.7× with the cut re-optimised and 2.5× at 0.779. At $f = 10\%$ moving the cut from 0.779 to 0.885 improves the limit by 22%. The counting figure of merit agrees with the limit up to $f = 10\%$; beyond it, it picks cuts whose B rests on a few raw events (0.945, $B = 8.9$) and is 12% worse.

4. **Nothing missing from the Λ-style chain changes its limit by much.** At $f = 15\%$ (NuWro, Poisson median): without the EXT estimate 39.8 against 37.6; with dirt at its bound ($+0.33 \pm 0.33$, the PeLEE dirt/overlay ratio at MuonCandidate) 43.5; with $g = 0$ 34.4; with B known exactly 36.5. The median n is an integer, so a small change in B moves the median limit in steps, either way. Hypana data and dirt would give the Λ-style chain an independent result with different systematics, not a better limit.

9 A global hypothesis fit

Status: not built. Its central idea, assigning the proton and pion jointly instead of one after the other, is realised by the pair scoring of `PidAssign.h` and of the Λ-style chain. The full four-role fit below remains a possible improvement; the reasoning and the measured ingredients are kept.

9.1 Motivation

The sequential selection identifies the muon, then the proton, then the pion, each conditioned only on what has already been taken, and never uses the fact that $p\pi^-$ must form a Λ and $\Lambda\gamma$ a Σ^0. The per-role PID templates show why this fails specifically on the pion: the muon, proton and photon each have a distinctive signature, while the π^- lies between the other three on every variable measured. The proposed fit relaxes the candidate pool (every PFP eligible for every role, recovering the 11.6% of π^- called showers), enumerates the $P(N, 4)$ assignments, scores each with a combined $-2 \ln L$ from the two mass constraints, the per-role PID templates, muon-at-vertex, the p–π DCA, a length-ordering prior and a soft-photon prior, and admits partial hypotheses for the 29.1% of signal with three PFPs. Perfect assignment given what is found is worth a factor 1.54 on $N \geq 4$ events.

9.2 Measured ingredients

Recorded in `HypFitParams.h` (1323 signal events): Λ mass constraint centred at 1.1234 GeV with $\sigma = 27.2$ MeV and Σ^0 at 1.1812 GeV with 45.4 MeV, from correctly assigned candidates. The Λ width predates the proton-momentum fix of Sec. 5, which narrows it to 7.8 MeV; it must be remeasured before use. The χ^2 must be centred on the measured means, not the PDG masses. The p–π DCA has median 0.35 cm (16–84%: 0.08–1.36 cm); the muon is the longest track 92.1% of the time; and 98.9% of PFPs carry both a track and a shower fit, so the relaxed pool needs no re-reconstruction.

variable	μ^+	p	π^-	γ
`trk_shr_score`	1.00	0.90	0.80	**0.11**
`trk_llrpid`	**0.96**	−0.29	0.14	0.56
`trk_pid_chipr`	180	**16.4**	96	192
`three_plane_dedx`	2.47	**10.3**	7.52	3.30
`rms_deflection`	0.01	0.01	0.01	**0.08**
`moliere_avg`	1.78	6.18	8.95	**13.3**
`trk_length`	**139**	12.8	10.8	14.2

Table 8: Per-role PID templates, medians on truth-matched PFPs. The π^- has no distinctive signature.

The mass-sculpting risk. Choosing the combination that best fits m_Λ sculpts a Λ-like peak out of combinatorics in background. If the fit is built, the masses are used for assignment only and kept out of the BDT, and the machinery is run on background alone to check for an induced peak.

10 Fake data by Poisson throw

The current "data" sample of the PeLEE chain is the GENIE Σ^0 QE sample, scaled by POT (until 2026-10-30, the NuWro signal file itself). That is a generator test of the signal efficiency, but not a closure test of the whole chain: it contains no background and no fluctuation. Read as data and corrected with NuWro's efficiency, it overestimates its own in-FV rate by about 8%, the generator dependence of the efficiency.

The fix is the Poisson-throw scheme of the CC1μ1π analysis (`macros/throw_perrun_*.C`): for each MC event with central-value weight w_{cv} and POT scale α, draw $n \sim \mathrm{Poisson}(w_{\mathrm{cv}}\alpha)$ and write the event n times with unit weight, across all samples. EXT, a data sample, enters at its scale directly. A `throw_sigma0.C` modelled on `throw_perrun_w.C`, one seed per sample, would provide it.

A throw from the same MC is a *statistical* closure test, not a model test. A model test needs a signal from a different generator in PeLEE format, and one now exists: the GENIE Σ^0 QE sample, 15 304 raw events at the operating point, ready to be injected into a throw of the NuWro-based prediction.

11 Framework port

The selection was to be ported into `xsec_analyzer` as a `SelectionBase` subclass to inherit its systematic machinery (flux multisims, GENIE multisims and unisims, detector variations, reinteraction), at a single analysis bin. **Status, 2026-10-19: not started.** No `Sigma0` class exists, there is no factory registration, and nine branches the selection uses are unbound in `Branches.hh` (details in `SIGMA0_FRAMEWORK_PLAN.md`).

The port is no longer on the critical path. The PeLEE ntuples carry the multisim weights themselves (`weightsGenie`, `weightsPPFX`, `weightsFlux`, `weightsReint`), the detector variations already run through the standalone selection (Sec. 3.6), and `Sensitivity.C` provides the limit machinery the framework lacks. A single-bin background covariance can be built in `SigmaSelection.C` directly.

12 Exposure: combined FHC+RHC later

The signal is $\bar{\nu}_\mu$ charged-current, so RHC is the sensitive mode. FHC carries a substantial wrong-sign $\bar{\nu}_\mu$ component (34.0% of the FHC flux, against 53.3% in RHC) and an exposure of 8.857 ×

10^{20} POT, so folding it in adds real signal. The CC1μ1π analysis has the machinery: per-run POT scaling, a combined `file_properties` and combined flux normalisation. It needs FHC overlay through the same selection, an FHC signal sample (or a justified reuse of the RHC one), FHC dirt and FHC EXT. At the purity of the PeLEE chain the background grows with the signal, so this is worth doing only once the systematics of Sec. 13 are in hand.

12.1 The FHC GENIE signal sample: weights, not a fixed GENIE

The FHC twin of the GENIE QE sample (`genie_uboone_NuMI_FHC_sigmazero_run1_2026`, `LambdaCCQE`, a primary Σ^0 selected) is being generated: 448 of 1200 chunks of 500 events on 2026-10-30. In FHC the Σ^0 comes only from the wrong-sign $\bar{\nu}_\mu$, since quasi-elastic hyperon production is $\bar{\nu}_\mu$-only, so it carries exactly the GENIE error of Sec. 2.6. Three ways to deal with it:

1. **Weights (adopted).** `HypXsecWeight.h` and `PolWeights.C` as for RHC. The weight depends only on E_ν and Q^2, not on the flux, so the table applies unchanged; it extends to 20 GeV and is clamped beyond. It costs no statistics (Sec. 2.6) and needs nothing regenerated. The polarisation weight needs the FHC hypana ntuples, which the overlay chain makes anyway.

2. **Rejection before the detector simulation.** Accepting each HepMC event with probability $w/w_{\max}$ gives an unweighted sample with the right distribution, but keeps only $\langle w\rangle/w_{\max} \approx 0.67/1.4 \approx 48\%$ of the Σ^0 the generation has already paid for. It gains nothing over weighting and is not worth it.

3. **Fix GENIE.** Replace `PaisQELLambdaPXSec::XSec` by the full contraction (or port NuWro's `Singh_Model`), regenerate the three $\bar{\nu}_\mu$ hyperon-QE splines on argon (cheap), and regenerate the sample. It is correct by construction, but it needs a private GENIE build inside the generation environment, a validation against the trace, and the full generation again. The polarisation weight is still needed, because Geant4 decays the Σ^0 and Λ isotropically whatever GENIE does.

The current FHC generation therefore continues unchanged and is weighted like RHC. The GENIE problem matters beyond this analysis: it moves GENIE's Λ prediction by $\times 1.26$, which is the model the MicroBooNE Λ measurement compared with, so it should be reported to the GENIE authors with the calculation of Sec. 2.6. A fixed GENIE is worth building if the collaboration wants unweighted official samples; it would replace the cross-section weight, not the polarisation weight.

13 Work plan

1. **Background systematics for the PeLEE chain (priority).** The expected limit grows $1.7\times$ from $f = 0$ to 15%, so this sets the result. Flux (`weightsPPFX`), cross section (`weightsGenie`) and reinteraction (`weightsReint`) multisims at the single bin, with the detector variations (7–10%, Sec. 3.6). Then a constraint from data: a fit of the BDT-score region where the signal is negligible, extrapolated to the signal region, which leaves only the shape uncertainty in f. The π^0 normalisation is the obvious target (Sec. 4.2).

2. **Choose the BDT cut from that budget** with the `Sensitivity.C` scan (0.885 at $f =$ 10%, 0.915 at 15%), and re-measure the detector variations there: the CV keeps fewer raw events at tighter cuts.

3. **Signal-model systematic.** Hyperon FSI is a factor ~ 4 between the generators; the GENIE normalisation is now settled by the QE sample (Sec. 2.4). The limit is quoted under

both models. The Λ cross-check against the published GENIE prediction should be redone on the QE sample, and so should the island tuning of the Λ-style chain.

4. **Closure and fake data.** `throw_sigma0.C`, all samples, with the GENIE Σ^0 QE sample as the injected alternative model (Sec. 10).

5. **Unblinding and the observed limit.** RHC beam-on data exist in PeLEE format (`custom_pelee_ntuples`). `Sensitivity.C` gives the observed limit once n is known; the absolute cross-section limit needs the integrated $\bar{\nu}_\mu$ flux and the argon target count in the FV.

6. **Λ-style chain as a cross-check.** Data and dirt through reco2 and HyperonProduction, as was done for EXT, would make it an independent measurement with different systematics. It does not improve the limit (Sec. 8), so it comes after the items above. The EXT factorisation of its island step is the open part of its background.

7. **Housekeeping.** At the next PeLEE retraining, add the input clamp of Sec. 3.3. The `Shower1MoliereAvg` histogram range is in radians while the variable is in degrees, so that control plot is empty at the BDT stage. (Done 2026-10-19: the hypana BDT now trains on 95% of each fold instead of 25% of the events, on the full background, AUC 0.824 $\rightarrow$ 0.834. Retrained on 2026-10-30 with the GENIE Σ^0 QE sample as signal: AUC 0.855, and at the operating point 0.554 $S = 17.1$, $B = 733$.)

8. **Combined exposure**, FHC+RHC (Sec. 12), and the framework port (Sec. 11) if it is still wanted.